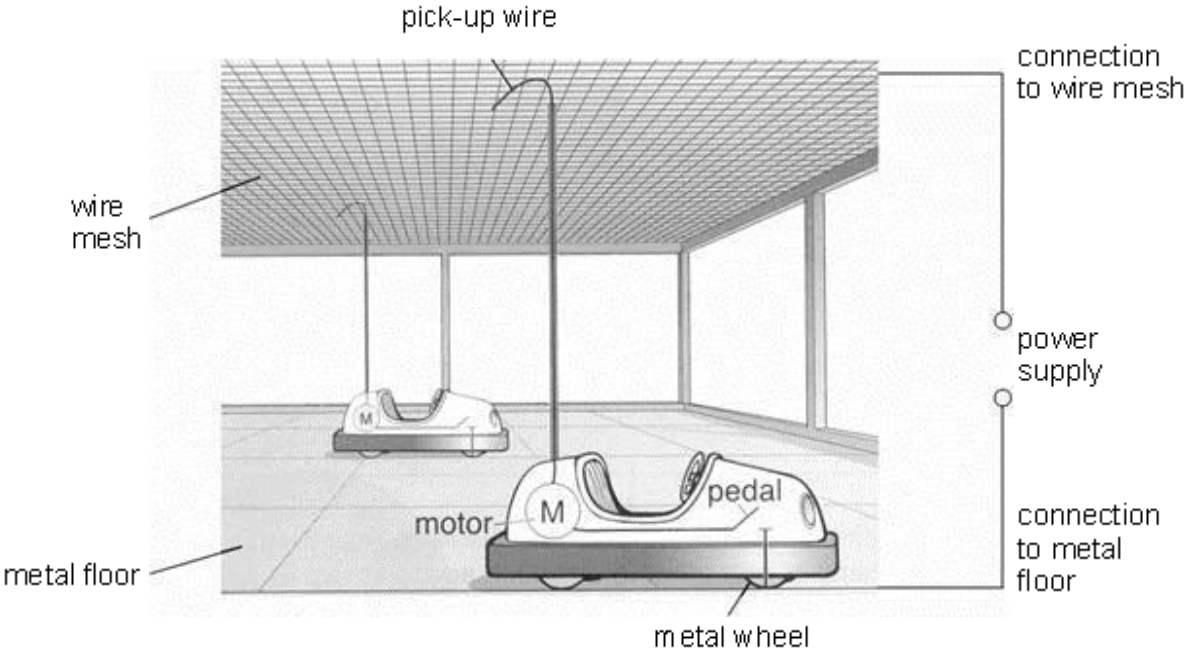
Q1. The diagram shows two dodgem cars at a fairground. The circuit symbols for the motor and pedal for each dodgem car are shown on the diagram.



(a) Complete the following sentence.

Each dodgem car is connected to the power supply through the

..... which is in contact with the wire

mesh, and through the ..... which is

in contact with the metal floor.

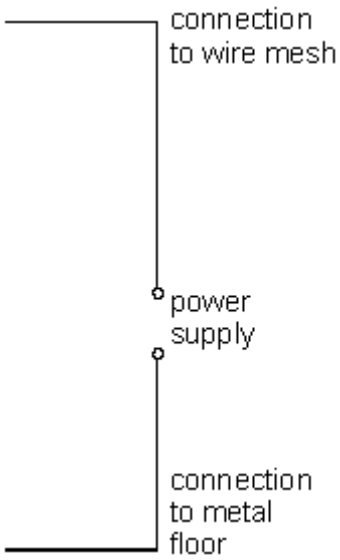
1 mark

(b) Dodgem cars are connected using parallel circuits.
Complete the circuit diagram below for the **two** dodgem cars.

Use **two** motor symbols, 

, and **two** switch symbols, ___/ ___.

The power supply for the circuit has been drawn for you.



2 marks

(c) Even when the power supply is switched on, the dodgem car will **not** move until the pedal is pressed. Give the reason for this.

.....

.....

1 mark

(d) A man looks after the dodgem cars during the rides.
Why does the man **not** get an electric shock as he walks across the metal floor?

.....

.....

1 mark

(e) During one ride, the two dodgem cars are running. The pick-up wire on one car snaps off.
Describe how this affects:

(i) the dodgem car with the broken pick-up wire;

.....

1 mark

(ii) the other dodgem car.

.....

1 mark

Maximum 7 marks

Mark schemes

Q1.

(a) pick-up wire

metal wheel

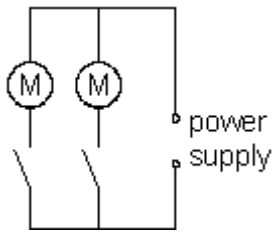
answers must be in the correct order

both answers are required for the mark

1 (L5)

(b) **One mark is for drawing the two motors connected in parallel.**
One mark is for drawing one switch in series with each motor.
Both marks should only be awarded if the circuit is correct.

•



the switches may be drawn either side of the motors

2 (L5)

(c) any **one** from

- it completes the circuit
- it acts as a switch

accept 'because the circuit is not complete'
'the pedal connects the motor' is insufficient
accept 'the pedal connects the motor to the power supply'

1 (L5)

(d) any **one** from

- he does not complete a circuit

accept 'the circuit is not complete'

- he does not connect the floor and wire mesh ceiling

accept 'he is not touching the ceiling'
*accept 'he wears trainers' **or** 'he has rubber shoes'*

1 (L5)

(e) (i) it stops

1 (L6)

(ii) it is not affected **or** it keeps going

accept 'it goes slightly faster'

1 (L6)

[7]